# Introduction

The recruitment process within companies is becoming increasingly complex due to the ever-growing competition for highly qualified specialists and the constant introduction of new technologies. Recruiters often face the problem of processing a large number of responses, sorting them, providing timely feedback and communication, as well as maintaining reports and statistics on vacancies. In addition, it is also necessary to make quick and timely decisions, taking into account the needs of the business and the needs of candidates. Despite all this, many companies in Uzbekistan, regardless of their size, still use outdated methods of storing information, such as spreadsheets, document templates that are not stored in the cloud, and verbal assignment of tasks. All this results in a slowdown and deterioration of the hiring process, significantly complicating both the recruiter's work and the candidate's experience in the process.

Working in this field, I have observed and noticed the consequences of not having such a system within a company, which is why I decided to create my own CRM system for HR teams. It will be designed specifically to meet all the needs of recruiters, namely to monitor and support candidates at all stages of the recruitment process. The system will help to structure and automate work processes and operations, improve communication with candidates and the transparency of all statuses and processes by storing all data about candidates and vacancies in one place and analysing the results to evaluate the effectiveness of recruitment.

The need for flexibility, rapid response and a high level of digitalisation in a company are modern labour market requirements, which emphasises the relevance of this project by optimising HR processes.

# Project Objectives

This section will discuss the main objectives of the project, identified based on the SMART method.

## Academic Objectives

- To apply the knowledge gained in classes over three years to create a real project, which fulfills the BISP requirements by April 2026.
- To conduct primary data collection to understand the need of the application on the market by February 2026 before the development phase.
- To analyze the market competitors and identify features needed during the research phase in January 2026.
- To successfully complete my studies and obtain a bachelor's degree, preferably with first-class honours.

**Commented [11]:** To produce a work which will fulfil BISP learning outcomes and allow me to …

## Business Objectives

- To develop an Applicant Tracking System (ATS) by the beginning of April, targeting Human Resources sphere and especially recruiters in Uzbekistan.
- To create an agreement with several companies for trial usage, including recruiting agencies in Uzbekistan in the first six months after the release.
- To achieve more that 60% of satisfaction rate from the clients in the first six months after deployment and speed up the hiring process for at least 20%.

## Technical Objectives

- To develop the ATS application using TypeScript programming language with basic functionalities: list of vacancies, candidate's profile, login page and page with statistics by the end of March 2026.
- To evaluate the effectiveness of the application by collecting user feedback through surveys and measuring key indicators such as user satisfaction, usability, and overall performance after deployment in May 2026.

## Personal Objectives

- To improve time management skills by meeting project deadlines and following a structured development schedule throughout the project.
- To enhance coding knowledge by successfully developing key features of the application during the development phase.
- To expand knowledge in the HRM field by conducting research and applying HR concepts within the project.

## SWOT Analysis

**Commented [12]:** I remember we agreed to move this to market analysis (or methodology business)

SWOT analysis helps to understand the strong and weak sides of the project, as well as possible opportunities and risks from outside.

### Strengths

One of the main strengths of this project is that it is designed for the local market in Uzbekistan. Many existing ATS systems are expensive and too complicated for small companies, but this system is more simple and affordable. This makes it easier for local businesses to start using it.

Another strength is the simple and clean design. The system is planned to be easy to use, so recruiters can quickly understand how everything works without long training. Also, all recruitment processes are in one place, which saves time and makes work more organized.

The project is also flexible, which means new features can be added later, like analytics or automation.

### Weaknesses

One weakness is that the system will not have many advanced features at the beginning. For example, it will not include AI tools or complex integrations like big platforms.

**Commented [13]:** But you have one, no?

Another problem is that the system is new, so people may not trust it at first. Many companies prefer to use well-known tools instead of trying something new.

There can also be technical issues, especially at the start. Since this is a student project, there may be problems with performance or stability.

### Opportunities

There are good opportunities for this project in Uzbekistan. Many companies still use simple tools like Excel or manual processes, so there is a need for better systems.

Another opportunity is support from IT Park and IT Park Ventures. They can provide funding, mentorship, and help with development.

In the future, the system can also expand to other countries in Central Asia, where similar problems exist.

### Threats

One of the main threats is strong competition from big platforms like SmartRecruiters and Recruitee. Even though they are expensive, they are popular and trusted.

Another threat is that new competitors can appear in the local market.

There are also risks related to technology, such as system bugs or data security problems, which can reduce user trust.

Finally, companies may not want to spend money on new systems if their budget is limited.

# Project Scope

This section will cover project scope including key project deliverables and technical requirements.
The project scope includes:

- To develop a web-based Applicant Tracking System (ATS) using modern frontend and backend technologies.
- To implement a relational database with full CRUD functionality using PostgreSQL as the database engine and Neon as the cloud database provider
- To implement a secure authentication system using JWT with role-based access control for different user types such as administrators, recruiters, and hiring managers

- To implement job management and candidate management modules with full create, read, update, and delete operations
- To implement an application pipeline that allows candidates to be connected to open jobs and moved through recruitment stages such as Applied, Screening, Interview, Offer, Hired, and Rejected
- To integrate a resume and document upload feature that allows candidate CVs and files to be uploaded into the ATS and stored securely through backend-controlled file handling
- To develop a user-friendly and professional interface inspired by SmartRecruiters, with responsive layouts, clear workflows, and bilingual support in English and Russian
- To implement search, filtering, sorting, and pagination functions for efficient job, candidate, and application management
- To implement candidate profile pages, job-specific candidate pipelines, and manual candidate creation linked to an open job
- To prepare the system for communication features such as interview notifications, messages, and email-based updates for recruitment activities
- To implement reporting and analytics features such as total jobs, total candidates, applications by stage, and hiring funnel statistics
- To test the system's functionality using appropriate software testing methods such as White Box Testing and Black Box Testing

The project scope excludes:

- To integrate an AI tool for assistance
- To have integrations with other Job Advertisement Platforms

## MoSCoW Prioritisation

### Must Have

- User authentication system using JWT with role-based access
- Job management module with full CRUD operations
- Candidate management module with full CRUD operations
- Application pipeline with stages
- Linking candidates to specific job vacancies
- Resume and document upload with secure storage
- Relational database implementation using PostgreSQL
- Basic user interface with clear navigation and workflows

#### Should Have

- Search, filtering, sorting, and pagination for candidates and jobs
- Candidate profile pages with structured information
- Job-specific pipelines for better organisation
- Responsive and user-friendly interface inspired by modern ATS systems

- Bilingual support
- Preparation for communication features

**Could Have**

- Advanced analytics and reporting
- Dashboard with visual statistics
- Improved UI/UX design and additional customisation options
- Activity logs and tracking of recruiter actions

**Won't Have**

- AI-based candidate screening or recommendations
- Integration with external job platforms (e.g. LinkedIn, job boards)
- Full CRM functionality for long-term candidate engagement
- Mobile application version

# Literature Review

Recruitment and HR practices are changing because of digital technologies. In the past, hiring was mostly manual and slow. Many tasks were done by hand and took a lot of time. Today, organisations use digital systems to improve recruitment. Two important systems are Applicant Tracking Systems (ATS) and Candidate Relationship Management (CRM). These systems help HR teams save time, manage candidates better, and improve hiring quality. These tools are used globally and are also becoming more common in Uzbekistan and Central Asia.

An Applicant Tracking System is software that helps organisations manage job applications. ATS stores candidate data, screens CVs, tracks applicants, and supports interview scheduling. It makes recruitment more structured and organised. According to Henderson (2024), ATS is widely used to handle large numbers of applications and to reduce manual work. Recruiterflow (2020) explains that ATS helps recruiters save time by automating repetitive tasks such as CV sorting and communication with candidates. This allows HR specialists to focus more on decision making.

Another benefit of ATS is better data management. ATS keeps all candidate information in one system. This improves reporting and transparency. PeopleHR (2026) states that ATS can improve the candidate experience because candidates receive faster feedback and clearer communication. ATS also helps HR teams measure important recruitment metrics such as time to hire and source of candidates.

CRM systems in recruitment work in a different way. Recruitment CRM focuses on building long term relationships with candidates. CRM systems allow recruiters to save contact information, send messages, and keep potential candidates engaged. Phenom (2025) explains

that CRM improves candidate engagement and helps organisations build talent pools for future hiring. This is useful when companies want to hire quickly later without starting the search from the beginning.

Many international HR articles explain that ATS and CRM work best together. ATS manages the recruitment process, while CRM manages communication and relationships. Geekflare (2025) states that ATS is designed for active candidates, while CRM is focused on passive candidates. When these systems are integrated, recruitment becomes more effective and strategic. TheTalent4U (2025) also notes that using both systems can reduce time to hire and improve hiring quality.

International experience shows clear benefits of digital recruitment systems. Recruiterflow (2020) reports that ATS reduces manual workload and improves productivity of HR teams. PeopleHR (2026) highlights that automated recruitment systems support better organisation and reduce human errors. CRM systems also help companies maintain contact with candidates and improve employer branding.

In Uzbekistan, digitalisation of HR is part of a wider national strategy. According to Times Kuwait (2026), the Uzbek government is introducing digital HR management systems to improve efficiency and transparency in public administration. These systems centralise employee data and reduce paperwork. Digital HR platforms also support analytics and reporting, which is similar to ATS functions such as tracking and data storage.

Local business sources show that HR practices in Uzbekistan are also changing. Revolution.uz (2024) reports that many companies follow global HR trends, including digital recruitment and automation. The article explains that companies use digital platforms to process CVs and evaluate candidates. This reduces hiring time and improves recruitment quality. These practices are close to ATS functionality even if companies do not use full ATS systems yet.

Spot.uz (2025) reports that the labour market in Uzbekistan faces staff shortages and growing demand for digital skills. Employers are increasingly interested in technology and automation in HR. Artificial intelligence and digital tools are becoming more common in recruitment. This creates strong reasons to adopt ATS and CRM systems in HR departments.

Academic research also supports HR digitalisation in Uzbekistan. The Central Asian Journal of Theoretical and Applied Science states that HR management in Uzbekistan is moving towards digital platforms and analytics. Recruitment technologies are part of this transformation. Digital HR systems help organisations follow international standards and improve efficiency.

However, some challenges still exist. Not all organisations fully use digital systems. Some companies only use basic features and still rely on manual methods. Lack of digital skills and experience can slow down adoption. Because of this, the full benefits of ATS and CRM are not always achieved.

In conclusion, international and Uzbek sources show that ATS and CRM systems provide many benefits for HR recruitment. These systems help automate hiring, improve candidate management, and support better decision making. In Uzbekistan and Central Asia, HR digitalisation is developing as part of economic and government reforms. Even though research is limited, existing sources show that digital recruitment tools are becoming more important and necessary for modern HR practices.

# Research Findings

Primary research consists of two parts: interviews with local and European recruiters, and a survey for other HR specialists such as HRBPs and HROps, who are less involved in daily work with ATS systems.

HRBPs (Human Resources Business Partners) usually work more with managers and focus on business goals rather than daily hiring tasks. They still use ATS systems, but mostly to check hiring progress, look at reports, and understand how recruitment is performing.

HROps (HR Operations specialists) focus on organizing HR processes and making sure everything works smoothly. They use ATS systems to manage data, support recruiters, and keep information updated, but they are not directly involved in hiring candidates.

So, while recruiters use ATS every day to work with candidates, HRBPs and HROps use it more for support, monitoring, and overall HR management.

Research consists of 5 parts, excluding introduction questions. The main goal is to receive information about current technology used and challenges with them, experience with ATS/CRM and feedback about them, pros and cons in these platforms and proposals for future development.

Primary research was conducted through three structured interviews with two recruiters and one team lead recruiter, each having between 2 to 5 years of experience in recruitment, mainly in IT and business-related roles. All participants described a similar recruitment cycle that starts from receiving a vacancy, sourcing candidates, screening CVs, conducting interviews, and finally hiring. At the moment, they use a combination of tools such as Excel, Google Sheets, LinkedIn, Telegram, and email, which makes the process less organised and more time-consuming. The respondents mentioned that candidate screening and tracking candidates across different stages take the most effort, especially when working with several vacancies at the same time.

In terms of challenges, all interviewees reported issues such as losing candidate information, missing important updates, and difficulties in managing recruitment stages like Screening and Interview. Two of them had experience using ATS systems such as SmartRecruiters and Workable. They found features like centralized candidate data, visual pipelines, and analytics useful, but also mentioned that these systems can be expensive, complex, and sometimes not

flexible enough for smaller teams. The team lead recruiter also highlighted the importance of having different access levels for users, which supports the idea of implementing role-based access control in the system.

All participants agreed that an ideal system should include features such as candidate and job management, a clear recruitment pipeline, CV and document storage, and easy search and filtering options. Automated communication, such as email notifications, was also considered very important for improving interaction with candidates. In addition, they showed interest in analytics features, especially metrics like time-to-hire, stage conversion rates, and overall recruitment performance. Finally, all respondents preferred cloud-based solutions for better data security and accessibility, and they emphasised the importance of a fast, user-friendly interface. These findings confirm the relevance of the proposed ATS system and support its planned features and functionality.

The second part of the primary research was conducted through an online survey, which received 25 responses from HR specialists. The participants included recruiters, HRBPs, HR coordinators, and team leads. The largest group of respondents were HR coordinators (40%), followed by recruiters (28%), HRBPs (24%), and team leads (8%). In terms of experience, most respondents had between 1-3 years of experience (36%), while others were distributed across less than 1 year (24%), 3-5 years (20%), and more than 5 years (20%). This shows that the data mainly represents junior and mid-level specialists who are actively involved in recruitment tasks.

The results show that many companies still use basic tools to manage candidates. Most respondents (72%) use Excel or Google Sheets, while 60% use internal databases and 52% rely on email. Only 24% of participants use ATS or CRM systems, which means that more advanced recruitment tools are not widely used due to their prices. Some respondents mentioned systems like SmartRecruiters, Workable, Recruitee, and Huntflow, but these are not common across all companies.

The main challenges identified in the survey were slow recruitment processes (68%), losing candidate information (52%), and lack of analytics (44%). Some respondents also mentioned data duplication and difficulties in managing the recruitment pipeline. These results show that current tools are not fully effective and create additional workload for recruiters.

The average satisfaction with the current recruitment process was 3.36 out of 5, which can be considered moderate. Most respondents selected ratings between 3 and 4, meaning that while the process works, it still has many areas that need improvement.

When looking at important features, the most requested one was a centralised candidate database (80%). Other important features included candidate tags and comments (64%), document upload (60%), and analytics dashboards (48%). Features like candidate interaction history and reminders were also considered useful. Automated email notifications were less important but still relevant for some users.

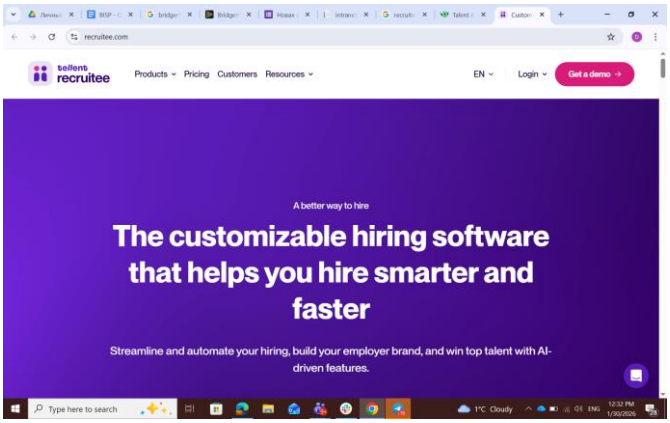
In terms of design, most respondents (64%) preferred a simple and minimalistic interface. Only a small number preferred more complex or detailed designs. Also, a mobile version was not considered very important, as most users prefer working from a laptop or computer.

Overall, the survey results support the interview findings and clearly show the need for a simple and well-structured ATS system. The results highlight problems with current tools and confirm that there is a demand for better data management, improved organisation, and basic analytics features. These findings strongly support the development of the proposed system.They show that many companies still rely on basic tools like Excel and email, which makes the recruitment process slower and less organized. The results also highlight key problems such as losing candidate data, lack of analytics, and difficulties in managing recruitment stages.At the same time, both interviews and survey results clearly show what users actually need: a simple system with a clear pipeline, centralized candidate database, and easy search and filtering. The preference for a clean and minimal interface also confirms that the system should focus on usability rather than complexity.

# Business Methodology

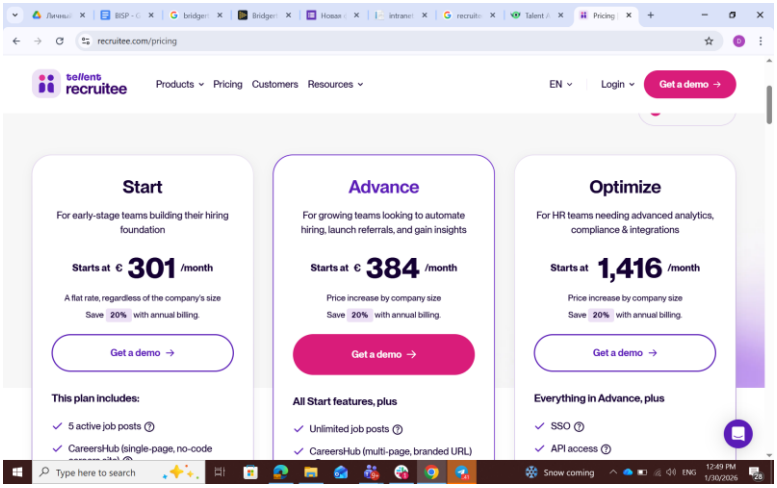
In order to better understand the CRM system development process specific for recruiters in Uzbekistan, it is necessary to look at the current global recruitment technology market and its existing solutions. The review is based on two principal platforms, Recruitee and SmartRecruiters, that not only showcase the most common features but also highlight the specific needs and drawbacks that will be referred to in the proposed MVP.

## Recruitee

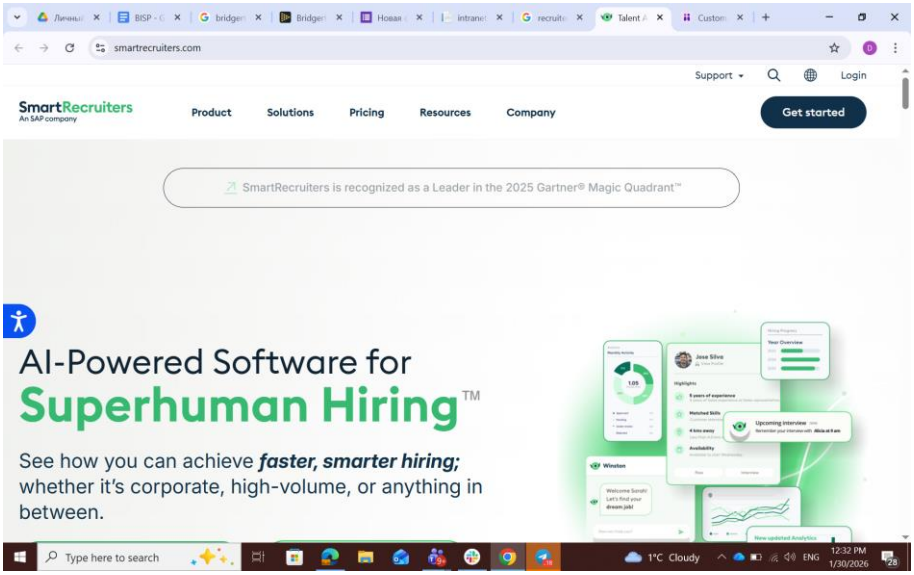



Recruitee is a web-based platform that focuses on the hiring process collaboration and primarily attracts small and medium-sized companies to use it. Utilizing a drag-and-drop kanban board, recruiters and hiring managers can organize candidates, follow hiring stages, plan interviews and communicate through internal notes. Customization is one of the most

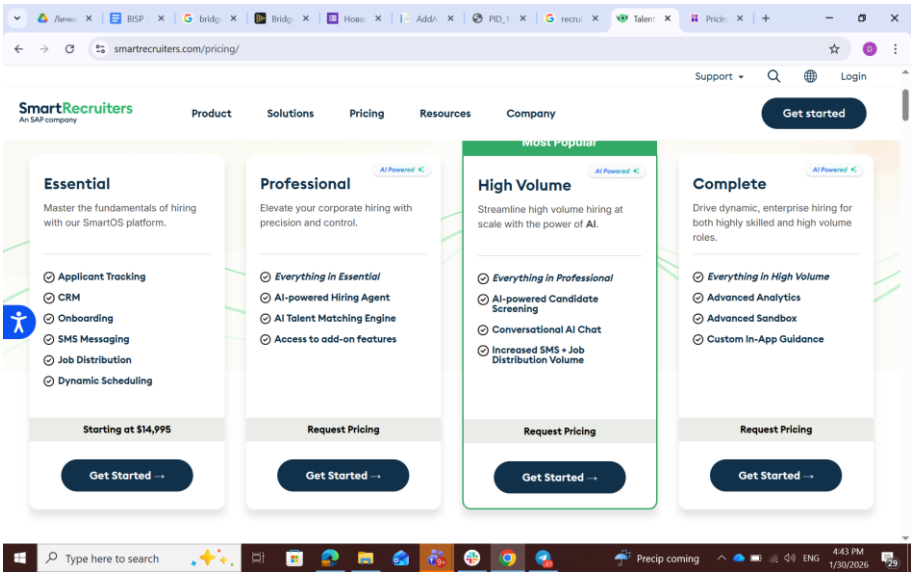
highlighted features. In Recruitee (2024), users drip pipelines, make structured evaluation templates, and even set up notifications automatically per vacancy. Recruitee has many features for team collaboration like commenting, mentioning colleagues in comments and sharing responsibilities for tasks. It also provides employers with branding tools, such as building a career site, and integrates with different job boards through multiposting. All these features make it an appealing solution for fast-growing businesses that need flexibility but do not want to deal with the complexity of enterprise systems.




## SmartRecruiters



SmartRecruiters is considered as a more sophisticated applicant tracking system (ATS) of an enterprise-grade. It comes with more features like hiring marketing, collaboration in the hiring process, candidate relationship management (CRM), and performance analytics. The system is equipped with the capability that is appropriate for large-scale use, and the big players like Visa, Bosch, and LinkedIn are the ones who are utilizing it (SmartRecruiters, 2012). The system has cutting-edge features such as AI screening, talent pools, requisition approval workflows, and ties with HRIS platforms. Although it is very powerful, the setup and operational intricacies render it unsuitable for small teams or organizations with minimal IT infrastructure.




To better understand how the proposed ATS system should be developed, a comparison of existing recruitment platforms such as Recruitee and SmartRecruiters was carried out. These platforms represent different levels of the market, where Recruitee is more suitable for small and medium-sized companies, while SmartRecruiters is designed for large organisations with more complex recruitment needs.

Recruitee is known for its simple and user-friendly interface, as well as flexible features such as a visual recruitment pipeline and team collaboration tools. It allows recruiters to organise candidates and track their progress easily. However, it has more limited analytics and automation compared to larger systems. In contrast, SmartRecruiters offers a wide range of advanced features, including AI-based screening, CRM functionality, and detailed analytics. At the same time, it is more complex to use and significantly more expensive, which makes it less suitable for smaller companies or local businesses in Uzbekistan.

Based on this comparison, the proposed ATS system aims to combine the most useful features of both platforms while avoiding their main disadvantages. The focus is on creating a system that is simple, affordable, and adapted to the needs of local recruiters. The system will include essential features such as candidate and job management, recruitment pipeline tracking, document storage, and analytics, without adding unnecessary complexity.

| Feature | Recruitee | SmartRecruiters | Proposed ATS System |
|---|---|---|---|
| Target Users | Small/medium companies | Large companies | Local companies |
| Complexity | Medium | High | Low to medium |
| Pricing | Around 384 eur/month | Around 1250 eur/month | N/A, will be calculated later |
| User Interface | Simple | Simple | Simple |
| Recruitment Pipeline | Visual | Advanced workflows | Visual |
| Candidate Management | Centralized data | Advanced system with CRM | CRUD functionality |
| Analytics | Basic | Advanced | Key metrics |
| Automation | Basic notifications | Integration with Teams | Email notifications (planed) |
| RBA | Yes | Yes | Yes |
| Documents Storage | Yes | Yes | Yes |
| Search and Filtering | Yes | Yes | Yes |
| Suitability for Uzbekistan | Affordable | Expensive | Planned to be affordable |

**Commented [14]:** For how many candidates per day? Or month?

**Commented [15]:** How do the differ? technically

**Commented [16]:** What is basic notification?

**Commented [17]:** And? Email eventually? What if candidate has not teams?

# Monetization Strategy

The monetization model for the Applicant Tracking System (ATS) will be based on a Software-as-a-Service (SaaS) model. This means that companies will pay a monthly subscription to use the system. This model is widely used in HR technologies because it provides stable income and allows the product to grow over time.

Based on the research of existing platforms, the average price of ATS systems is around €400-600 per month (Recruitee, 2024; SmartRecruiters, 2012) . However, prices depend on features and system complexity. For example, Recruitee offers a more affordable solution starting from around €384 per month, which is suitable for small and medium-sized companies (Recruitee, 2024). In contrast, SmartRecruiters is designed for large companies and starts from approximately €1,250 per month (SmartRecruiters, 2012) .

Considering the economic situation in Uzbekistan, it is important to make the system more affordable. The average monthly salary in Uzbekistan is around 5.7-6.3 million soums (approximately $400-500), which shows that companies have limited budgets for software solutions (RemotePeople, 2026; National Statistics Committee, 2025). Therefore, this project will offer lower and more flexible pricing depending on company size and features. This approach will help attract more users and increase adoption in the local market.

In addition to subscriptions, the system may use a freemium model. Basic features such as limited job postings and candidate storage will be free, while advanced features like analytics and automation will require payment. This helps attract users and later convert them into paying customers (Recruitee, 2024).

Another important opportunity is sponsorship and funding. IT Park Uzbekistan supports startups by providing infrastructure, tax benefits, and development programs (IT Park Uzbekistan, 2026). In addition, IT Park Ventures is a venture fund created to invest in startups and help them grow. It provides funding, mentorship, and support at different stages, from idea to scaling (IT Park Ventures, 2025).

From the cost perspective, one of the main expenses is hosting and infrastructure. For example, cloud providers such as DigitalOcean offer basic servers starting from around $6-12 per month, which is suitable for small applications (DigitalOcean, 2025). For a more complete setup with backend, database, and storage, the cost can reach around $200 per month (DigitalOcean, 2025). Another option is Amazon Web Services (AWS), which uses a pay-as-you-go model, meaning companies only pay for the resources they use, but total costs may vary depending on usage (AWS, 2026).

Database costs are also important. For example, managed MySQL databases on DigitalOcean start from around $15 per month for basic usage (DigitalOcean, 2025). Similarly, services like AWS RDS or Google Cloud SQL can start from approximately $11-12 per month for small instances, but prices increase depending on performance and storage (Bytebase, 2025). Storage costs are usually calculated per gigabyte and can be around $0.10-0.17 per GB per month depending on the provider (Bytebase, 2025).

Another major cost is the development team. Salaries in Uzbekistan are relatively lower compared to global markets, which makes development more affordable. For example, the average monthly salary in Uzbekistan is around $400-500 (RemotePeople, 2026), while software developers can earn around $1,500-2,000 per month depending on experience (HireTruss, 2026; Glassdoor, 2026). This means that a small development team may cost

around $1,500 to $4,000 per month. Additional costs include domain registration, email services, and marketing, which are necessary for promoting the system and attracting users.

In conclusion, the monetization strategy is based on a subscription model supported by additional revenue options and funding opportunities. By offering lower prices compared to global platforms like SmartRecruiters and Recruitee, and by using support from IT Park and IT Park Ventures, the system has strong potential to grow and succeed in the Uzbek market.

### Subscription Price Calculation

To define a realistic subscription price, it is important to consider both operational costs and basic development expenses, including salaries.

At the early stage, the system has the following monthly costs:

- Hosting and database: around $70-100 per month
- OpenAI API usage: around $10-20 per month
- Total infrastructure cost: approximately $100 per month

In addition, even though this is a student project, it is important to consider estimated development costs to understand the real business value. For example, a small team may include one developer with an average salary of around $800-1,000 per month in Uzbekistan.

This means the total monthly cost can be estimated as:

- Infrastructure: $100
- Development (salary): $900 for one full-stack developer (at least two needed)
- Total: $1,900 per month

For one year, the total cost would be:

- $1,900 × 12 = $22,800 per year

To make the system sustainable and profitable, the project aims to generate around $300-400 per month in the first year. This means the system will not fully cover all development costs at the beginning, but it will cover operational costs and part of the investment.

Instead of changing the price depending on the number of clients, a fixed subscription price is more practical and easier to manage. Therefore, the system can set a fixed price of around $600 per month per company.

With this pricing:

- 3 clients → 3 × $600 = $1800 per month
- 4 clients → 4 × $600 = $2400 per month

This allows the system to reach the target income while keeping the price affordable for small and medium-sized companies.

# IT Methodology

## Development Process

The system will be developed using a simplified Agile methodology, structured into three short Phases:

- Phase 1 shifts its focus to the primary setup components. The maintenance of vacancy records, candidate profile management, and authentication systems are among the crucial elements covered in the system documentation.
- Phase 2 carries on the work by putting in place a pipeline system in the kanban style. Activity logging, dashboard elements, and data export capabilities are the three primary features of the system. By making changes to enhance usability,
- Phase 3 brings the process to a close. Performance optimization, final testing, user feedback analysis, and acceptance criteria verification are all necessary steps in the process.
- Agile was chosen because of its adaptability, quick feedback, and suitability for projects with changing requirements and design learning over time.

**Commented [18]:** Any tool for project management?

## User Platform

The system will be created with an emphasis on user-centered design, giving usability and clarity top priority: Each component, including the recruitment pipeline, job postings, and candidate profile, will be depicted by wireframes. The user flows will be created so that recruiters can complete necessary tasks like updating a candidate's status or logging an interaction with three clicks or less.

## Tools, Libraries, and Technologies Used

### Frontend

React was used to build the user interface. It helps to create reusable components, which makes the system easier to manage and update. This is very useful for an ATS system because it includes many dynamic elements like dashboards, tables, and candidate profiles.

Vite was used as a development tool. It makes the application run faster during development and helps to see changes instantly without reloading the whole page.

Tailwind CSS was used for styling. It allows quick design using ready-made classes and helps keep the interface clean and consistent.

React Router DOM was used for navigation between pages like dashboard, candidates, jobs, and reports. It allows smooth transitions without reloading the page.

Axios was used to send requests between the frontend and backend. It makes it easier to work with APIs and handle data.

### Backend

Node.js was used as the backend environment. It allows using JavaScript on the server side, which makes development easier because the same language is used everywhere.

Express.js was used to build the backend structure and APIs. It helps organize routes and handle requests in a simple way.

Prisma ORM was used to work with the database. It makes it easier to define data models and manage relationships between different parts of the system like users, jobs, and candidates.

### Database

PostgreSQL was used as the main database. It is reliable and works well with structured data, which is important for systems like ATS.

Neon was used as a cloud database provider. It allows easy setup and management of PostgreSQL without needing to install it locally.

### Authentication and Security

JWT (JSON Web Tokens) was used for authentication. It allows users to log in securely and keeps sessions simple.

bcryptjs was used to hash passwords before storing them, which improves security.

Role-based access control was added to limit access based on user roles like admin, recruiter, or hiring manager.

### File Handling

Multer was used to upload files such as CVs. This is an important feature for storing candidate documents.

**Commented [19]:** What is this?

### Email and Communication

Nodemailer was used to send emails. It helps with communication like interview invitations and updates.

### AI Integration

OpenAI API was used to analyze resumes. It helps extract useful information like skills and experience, which makes recruiters' work easier.

**Commented [110]:** How good is it from ethical or privacy/security points of view?

### Development Tools

dotenv was used to store important data like passwords and keys securely.

Morgan was used to log backend requests, which helps with debugging.

CORS was used to allow frontend and backend to communicate safely.

**Commented [111]:** What are they? What are alternatives?

### Deployment Platforms

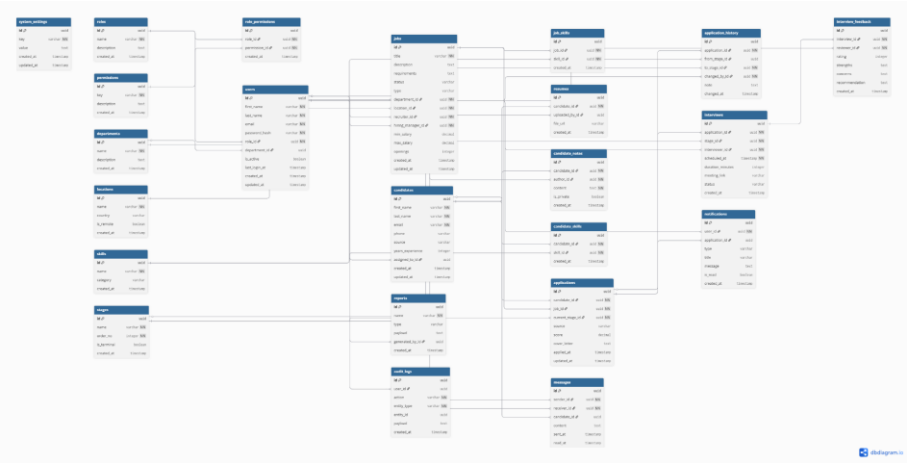
Render was used to deploy the backend.

Vercel was used to deploy the frontend. It works well with React and makes deployment simple.

**Commented [112]:** Why not both on same platform?

## Database Scheme

**Commented [113]:** It is called ERD



UML? Use case? Architecture?

# Testing

First, black-box testing was used to check how the system behaves from the user's side. This included testing things like logging in, creating jobs, managing candidates, moving them through pipeline stages, scheduling interviews, sending emails, and using the AI resume analysis feature. The idea was to see if everything works as expected from a normal user's perspective.

Then, white-box testing was used to check the internal logic of the system. This focused more on the backend, including things like authentication, role-based access, error handling, JWT logic, and helper functions. Automated tests were used here to make sure these parts are working properly.

System testing was also done to see how the whole application works together. This included checking how the frontend, backend, database, and external services like Gmail and OpenAI interact with each other. It was important to make sure everything is connected and runs smoothly as one system.

For automation, Node.js built-in testing tools were used for backend testing. At the same time, manual testing was done for the user interface, including checking workflows, permissions for different roles, and how the dashboard updates in real time.

**Commented [114]:** What are the results? How many tests? All successful?

Overall, using both manual and automated testing helped make sure that the system is not only technically correct but also practical and easy to use for real users.

# Challenges Faced Over Development

During the development of the project, several challenges were faced. One of the main challenges was understanding how to structure the system correctly, especially at the beginning. Since the project includes both frontend and backend, it was sometimes difficult to manage the connection between different parts of the system.

Another challenge was working with the database and relationships between entities such as users, jobs, and candidates. At some stages, issues appeared with data linking and updating records correctly. Fixing these problems required additional time and testing.

Integration with external services such as email and AI tools was also challenging. For example, setting up email functionality and making sure messages are sent correctly took several attempts. Similarly, working with the OpenAI API required understanding how to process and display results properly.

Time management was another difficulty. Balancing development, research, and writing the report at the same time required careful planning. Some features took longer than expected, which made it necessary to adjust the timeline.

# Timeline

The project was carried out over an 8-month period, starting in September 2025 and finishing at the end of April 2026.

- **September 2025** - Initial planning
  Defining project idea, objectives, and overall direction
- **October - November 2025** - Research phase
  Literature review, competitor analysis, and understanding ATS systems
- **December 2025 - January 2026** - Primary research
  Conducting interviews and survey, analysing results
- **February 2026** - System design
  Planning system architecture, database structure, and UI design
- **February - March 2026** - Development phase
  Implementing core features such as authentication, job and candidate management, and pipeline
- **March - April 2026** - Integration
  Adding email functionality, resume upload, and additional features
- **April 2026** - Testing and final improvements
  Fixing bugs, improving performance, and preparing final version

**Commented [115]:** Detailed gantt chart will be much better

# Critical Self-Evaluation

Overall, the project can be considered mostly successful, as it meets the main objectives and provides a working ATS system. The system includes key features such as candidate management, job tracking, and recruitment pipeline, which are useful for real HR processes.

One of the strengths of the project is the clear structure and focus on user needs. The system was designed based on real research, which helped ensure that the features are relevant and practical.

However, there are also areas for improvement. Some parts of the system could be more optimized, especially performance and scalability. In addition, more advanced features could be implemented to make the system more competitive.

From a personal perspective, this project helped me to improve technical skills, especially in full-stack development, and provided a better understanding of how real systems are built.

# Limitations

This project has several limitations. First, the system does not include advanced features such as AI-based candidate recommendations or full automation of recruitment processes.

Second, the system does not support integration with external job platforms such as LinkedIn or job boards. This limits its functionality compared to larger ATS systems.

Another limitation is scalability. The system was developed as a prototype, so it may require additional improvements to handle a large number of users or data.

# Future Improvements Plan

There are several ways the system can be improved in the future. One important improvement is adding more advanced analytics, such as detailed recruitment reports and performance tracking.

Another improvement is integrating the system with external job platforms and services, which would make the recruitment process more automated.

The system can also be improved by adding AI features, such as candidate recommendations and smarter resume analysis.

In terms of user experience, further improvements can be made to the interface and usability, based on user feedback.

Finally, the system can be scaled and optimized to support more users and larger datasets, making it suitable for real business use.

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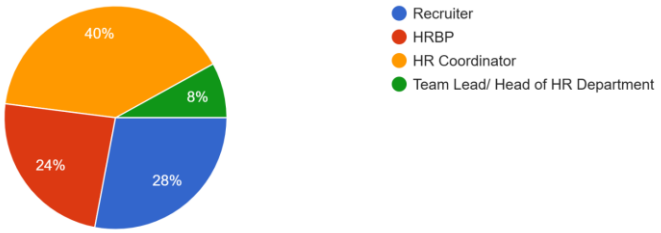
# Appendix

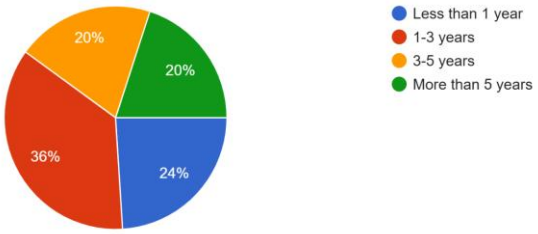
**Commented [116]:** Give a subtitle -what is it?

What is your current role?\ На какой должности вф сейчас работаете?
25 ответов



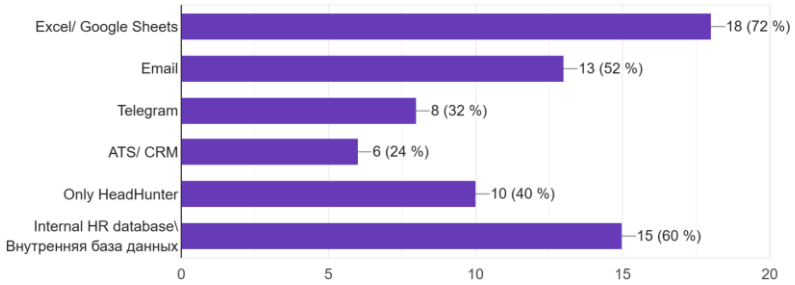
How many years of experience do you have in HR?\ Сколько лет вы уже работаете в HR?

25 ответов




What tools do you currently use to manage candidates? (select all that apply)\ Какими програмами вы пользуетесь во время работы?

25 ответов



Please name ATS/CRM system you are using\ Перечислите названия ATS/CRM систем, которые вы используете

9 ответов

SmartRecruiters

Мы пользуемся только HH, так все данные по кандидатам храним в Excel таблицах

Есть опыт с HuntFlow и Recruitee

Huntflow

Workable

Recruitee

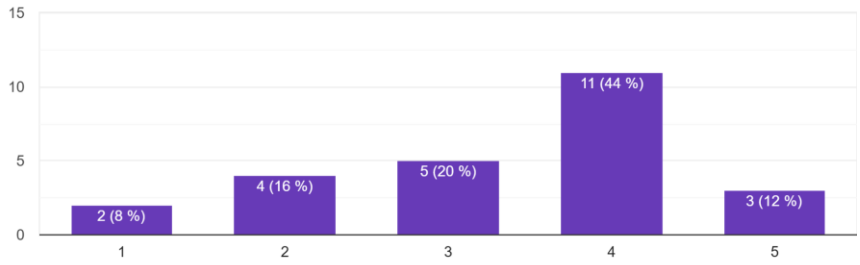
Другая АТС система, не для рекрутинга, а именно для внутренних сотрудников

У меня есть опыт работы с Recruitee, но это было на прошлой работе

SmartRecruiters, GreenHouse

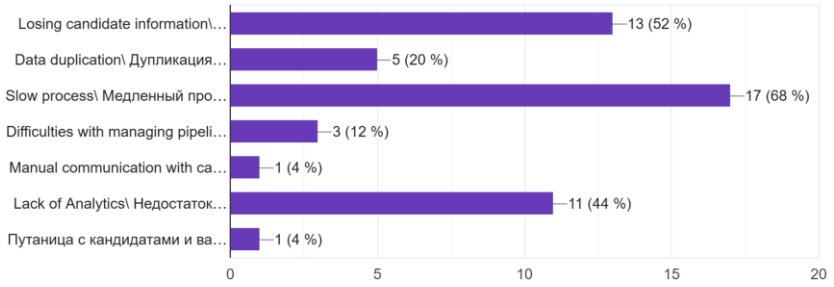
How satisfied are you with your current recruitment process? Насколько вы удовлетворены процессов рекрутинга у вас в компании?

25 ответов



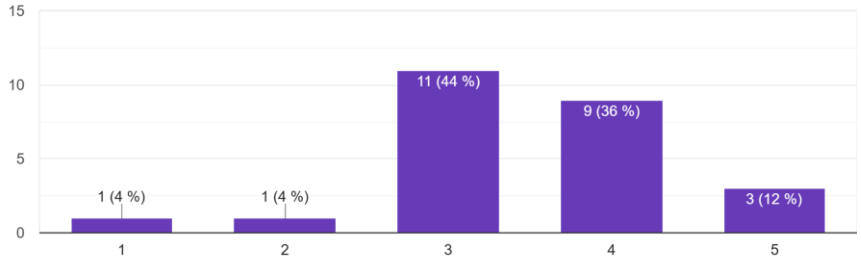
What challenges do you face most often? (Select all that apply)\ С какими проблемами вы чаще всего сталкиваетесь?

25 ответов



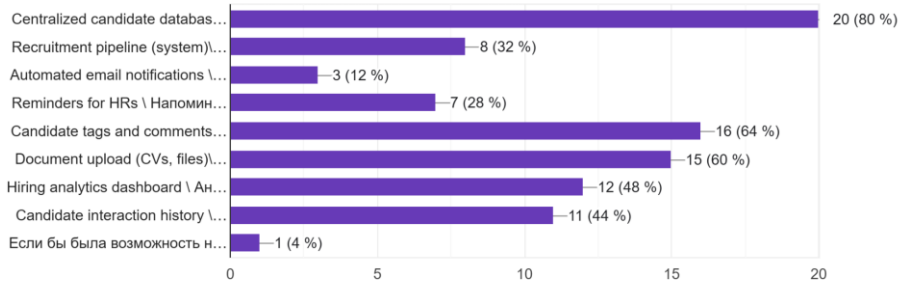

How frequently do you experience these challenges? \ Как часто вы сталкиваетесь с этими проблемами?

25 ответов



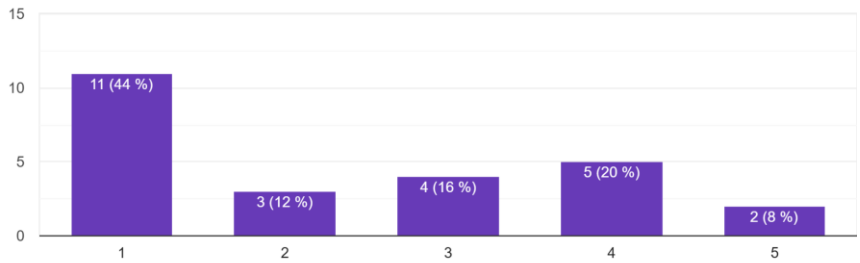

Which of these features are important for you? \ Насколько вам важны эти функции в приложении?

25 ответов



What type of design do you prefer in ATS system? \ Какой дизайн вы бы предпочли видеть?

25 ответов

How important is having a mobile-friendly version? \ Насколько важно существование мобильной версии приложения?

25 ответов